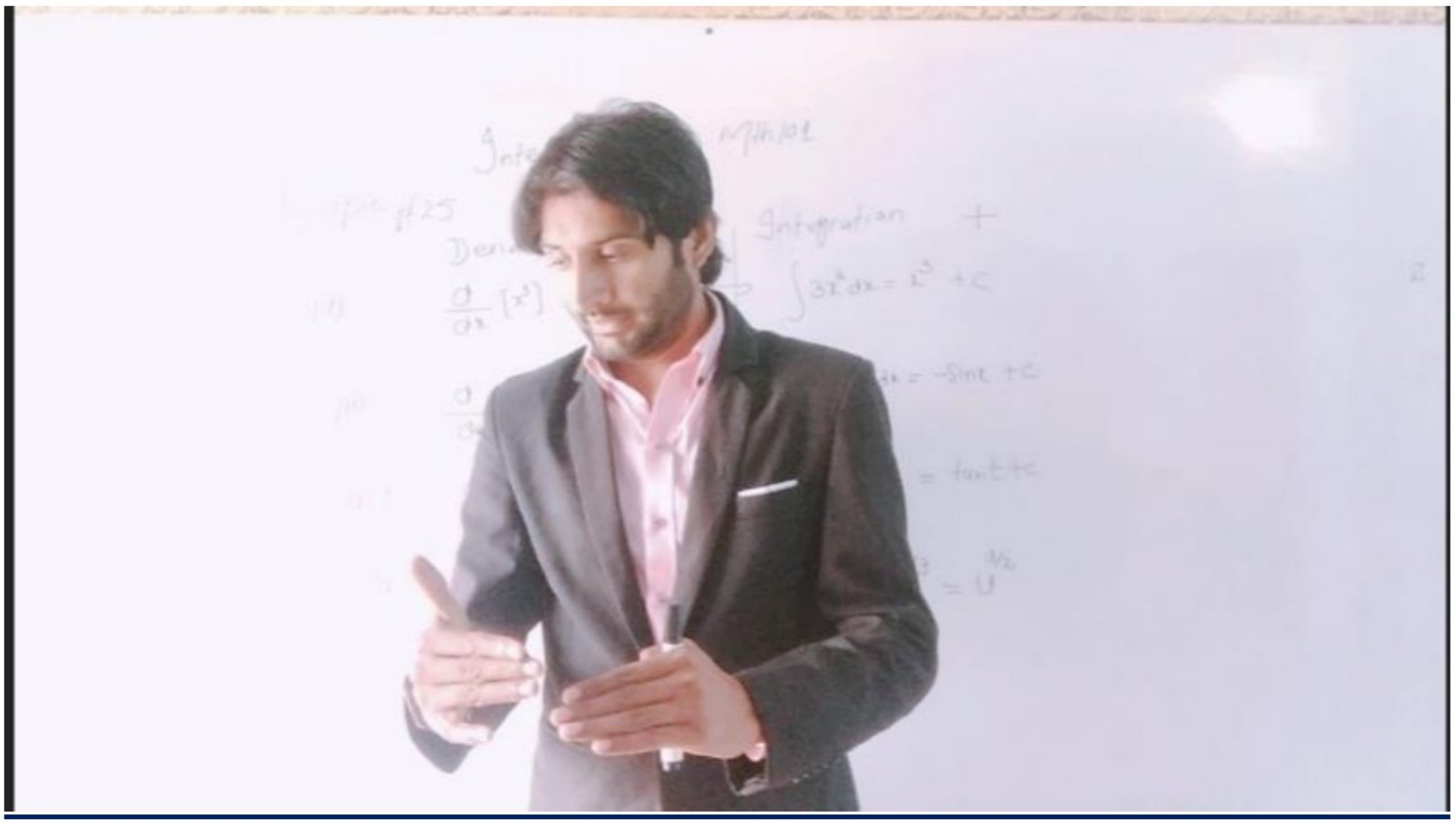
Mth603 MidTerm Quiz Preparation Fall 2021

YouTube:Life Star Academy

By Sir Ray Afzal



Question No:01

While solving a system of linear equations, which of the following approach is economical for the computer memory?

Direct

Iterative (Page 69)

Analytical

Graphical

Question No:02

The basic idea of relaxation method is to reduce the largest

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residual to ...

One
Two
Zero (Page 83)
None of the given choice

Question No:03

The Jacobi's method is a method of solving a matrix equation on a matrix that has no zeros along its_____.

Main diagonal (Page 104)
last column
last row
first row

Question No:04

If A is a nxn triangular matrix (upper triangular, lower triangular) or diagonal matrix,the eigenvalues of A are the diagonal entries of A.

TRUE Click here for detail
FALSE

Question No:05

A 3 x 3 identity matrix have three and different Eigen values.

TRUE Click here for detail
FALSE

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Question No:06

Which of the following is a reason due to which the LU decomposition of the system of linear equations; x+y = 1, x+y =2 is not possible?

Associated coefficient matrix is singular
All values of l's and u's can't be evaluated
Determinant of coefficient matrix is zero
All are equivalent

Question No:07

Gauss - Jordan Method is similar to …

Gauss–Seidel method
Iteration's method
Relaxation Method
Gaussian elimination method (Page 95)

Question No:08

While using Relaxation method, which of the following is the largest Residual for 1st iteration on the system; 2x+3y = 1, 3x +2y = - 4 ?

-4
3
2
1

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Question No:09

Gauss–Seidel method is also known as method of …

Successive displacement (Page 263)

Iterations

False position

None of the given choices

Question No:10

Jacobi's Method is a/an…

Iterative method (Page 69)

Direct method

Question No:11

The characteristics polynomial of a 3x 3 identity matrix is _____, if x is the eigen values of the given 3 x 3 identity matrix. where symbol ^ shows power.

(x-1)^3

(x+1)^3

x^3-1

x^3+1

Question No:12

The can be used only to find the eigenvalue of A that is largest in absolute value—we call this eigenvalue the dominant eigenvalue of A.

TRUE

FALSE

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Question No:13

In ... method, a system is reduced to an equivalent diagonal form using elementary transformations.

Jacobi's
Gauss-Seidel
Relaxation
Gaussian elimination (Page 262)

Question No:14

The linear equation: 2x+0y-2=0 has ------- solution/solutions.

Unique (Page 48)
no solution
infinite many
finite many

Question No:15

Under elimination methods, we consider, Gaussian elimination and ...methods.

Gauss-Seidel
Jacobi
Gauss-Jordan elimination (Page 48)
None of the given choices

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Question No:16

Which of the following method is not an iterative method?

Jacobi's method
Gauss-Seidel method
Relaxation methods
Gauss-Jordan elimination method

Question No:17

An eigenvector V is said to be normalized if the coordinate of largest magnitude is equal to zero.

TRUE
FALSE (Page 97)

Question No:18

Exact solution of 2/3 is not exists.

TRUE
FALSE

Question No:19

When the condition of diagonal dominance becomes true in Jacobi's Method.Then its means that the method is ...

Stable

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Unstable

Convergent (Page 70)

Divergent

Question No:20

Gauss–Seidel method is similar to …

Iteration's method

Regula-Falsi method

Jacobi's method

None of the given choices (Page 263)

Question No:21

Sparse matrices arise in computing the numerical solution of…

Ordinary differential equations

Partial differential equations (Page 69)

Linear differential equations

Non-linear differential equations

Question No:22

While solving by Gauss-Seidel method, which of the following is the first Iterative solution for the system; x-2y =1, x+4y=4 ?

(1, 0.75)

(0 ,0)

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(1,0)
(0,1)

Question No:23

While solving a system of linear equations by Gauss Jordon Method, after all the elementary row operations if there lefts also zeros on the main diagonal then which of the is true about the system?

System may have unique solutions
System has no solution
System may have multiple numbers of finite solutions
System may have infinite many solutions

Question No:24

Numerical methods for finding the solution of the system of equations are classified as direct and... methods

Indirect
Iterative (Page 48)
Jacobi
None of the given choices

Question No:25

If the Relaxation method is applied on the system; 2x+3y = 1, 3x +2y = - 4, then largest residual in 1st iteration will reduce to -------

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zero

4

-1

-1

Question No:27

While using Relaxation method, which of the following is the Residuals for 1st iteration on the system; 2x+3y = 1, 3x +2y =4 ?

(2,3)

(3,-2)

(-2,3)

(1,4)

Question No:28

If the order of coefficient matrix corresponding to system of linear equations is *33 then which of the following will be the orders of its decomposed matrices; 'L' and 'U'?*

Order of 'L' = *31*, Order of 'U' = *13*

Order of 'L' = *32*, Order of 'U' = 2*3

Order of 'L' = *33*, *Order of 'U'* = *33*

Order of 'L' = *34*, *Order of 'U'* = *43*

Question No:29

While solving the system; x–2y = 1, x+4y = 4 by Gauss-Seidel method, which of the following ordering is feasible to have good approximate solution?

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x+4y = 1, x-2y = 4
x+2y = 1, x- 4y =4
x+4y = 4, x–2y = 1
no need to reordering

Question No:29

Full pivoting, in fact, is more …than the partial pivoting.

Easiest
Complicated (Page 51)

Question No:30

For the equation
$x^3+3x-1=0$ **the root of the equation lies in the interval…**

- (1, 3)
- (1, 2)
- **(0, 1)**
- (1, 2)

Question No:31

…lies in the category of iterative method.

- Bisection Method
- Regula Falsi Method
- Secant Method
- **all of the given choices (Page 8)**

Question No:32

If n x n matrices A and B are similar, then they have the

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different eigenvalues (with the same multiplicities).

True

False

Question No:33

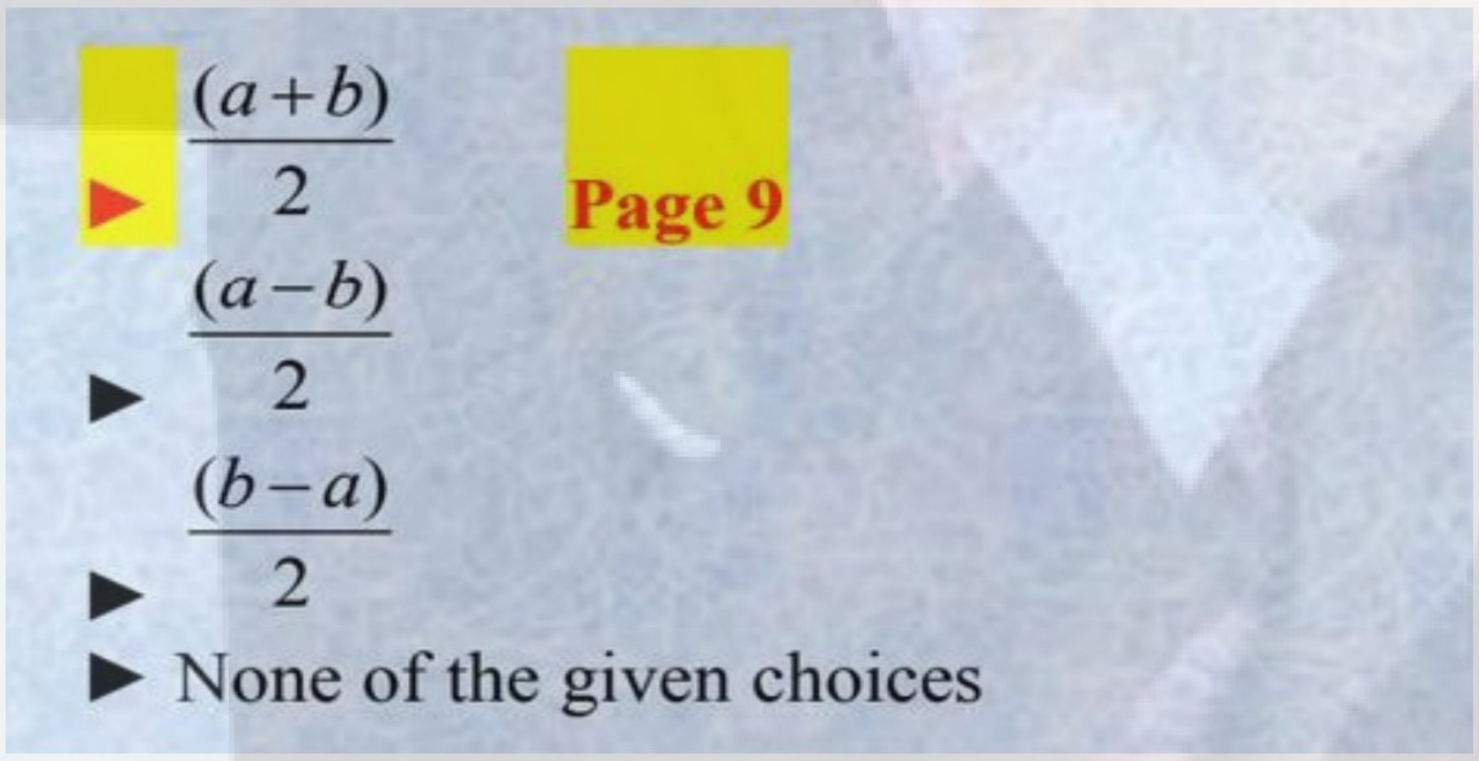
The Jacobi's method is a method of solving a matrix equation on a matrix that has ____zeros along its main diagonal.

No Click here for detail

At least one

Question No:34

If the root of the given equation lies between a and b, then the first approximation to the root of the equation by bisection method is ...



Question No:35

To apply Simpson's 3/8 rule, the number of intervals in the following must be

► 10

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► 11
► **12**
► 13

Question No:36

The Gauss-Seidel method is applicable to strictly diagonally dominant or symmetric_____ definite matrices A.

Positive Click here for detail
negative

Question No:37

Differences methods find the _____ solution of the system.

Numerical
Analytical

Question No:38

To apply Simpson's 1/3 rule, the number of intervals in the following must be
► 2 (Simpson's 1/3 rule must use an even number of elements')
► 3
► 5
► 7

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Question No:39

Bisection and false position methods are also known as bracketing method and are always

Divergent

Convergent (Page 26)

Question No:40

The Inverse of a matrix can only be found if the matrix is.....

Singular

None Singular: Every square non-singular matrix will have an inverse.

Scalar

Diagonal

Question No:41

In interpolation is used to represent the δ Forward difference Δ

Central difference (Page 117)

Backward difference

Question No:42

The base of the decimal system is _____

10

0

2

8

None of the above.

Question No:43

Bisection method is ... method

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► Open Method
► **Bracketing Method (page 26)**

Question No:44

A 3 x 3 identity matrix have three and _____Eigen values.

Same

Different

Question No:45

Eigenvalues of a symmetric matrix are all _____ .

Real (page 104)

complex

zero

positive

Question No:46

The Jacobi iteration converges, if A is strictly diagonally dominant.

TRUE (Page 69)

FALSE

Question No:47

Below are all the finite difference methods EXCEPT _____.

Jacobi's method

Newton's backward difference method

stilling formula

Forward difference method

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Question No:48

Two matrices with the same characteristic polynomial need not be similar.

TRUE

FALSE

Question No:49

The determinant of a diagonal matrix is the product of the diagonal elements.

True

False

Question No:50

The Gauss-Seidel method is applicable to strictly diagonally dominant or symmetric positive definite matrices A.

True

False

Question No:51

The determinant of a _____ matrix is the product of the diagonal elements.

Diagonal Page No.70

Upper triangular

Lower triangular

Scalar

Question No:52

For differences methods we require the set of values.

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True

False

Question No:53

If x is an Eigen value corresponding to Eigen value of V of a matrix A. If a is any constant, then x – a is an Eigen value corresponding to Eigen vector V is an of the matrix A - a I.

True

False

Question No:54

Central difference method seems to be giving a better approximation, however it requires more computations.

True

False

Question No:55

Iterative algorithms can be more rapid than direct methods.

True

False

Question No:56

Central Difference method is the finite difference method.

True

False

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Question No:57

Back substitution procedure is used in …

Gaussian Elimination Method

Jacobi's method

Gauss-Seidel method

None of the given choices

Question No:58

The Jacobi's method is a method of solving a matrix equation on a matrix that has no zeros along its main diagonal.

True

False

Question No:59

Power method is applicable if the Eigen vectors corresponding to Eigen values are linearly independent.

True (Page102)

False

Question No:60

Power method is applicable if the Eigen values are _____.

Real and distinct (Page102)

real and equal

positive and distinct

negative and distinct

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Question No:61

Simpson's rule is a numerical method that approximates the value of a definite integral by using polynomials.

Quadratic (Page174)

Linear

Cubic

Quartic

Question No:62

In Simpson's Rule, we use parabolas to approximating each part of the curve. This proves to be very efficient as compared to Trapezoidal rule.

True Click here for detail

False

Question No:63

The predictor-corrector method an implicit method. (multi-step methods)

True (Page 212)

False

Question No:64

Generally, Adams methods are superior if output at many points is needed.

True

False

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Question No:65

The Trapezoidal rule is a numerical method that approximates the value of a._____.

Indefinite integral

Definite integral (Page176)

Improper integral

Function

Question No:66

The need of numerical integration arises for evaluating the definite integral of a function that has no explicit _____ or whose anti derivative is not easy to obtain.

Ant derivative

Derivatives.

Question No:67

An indefinite integral may _____ in the sense that the limit defining it may not exist.

Diverge

Converge

Question No:68

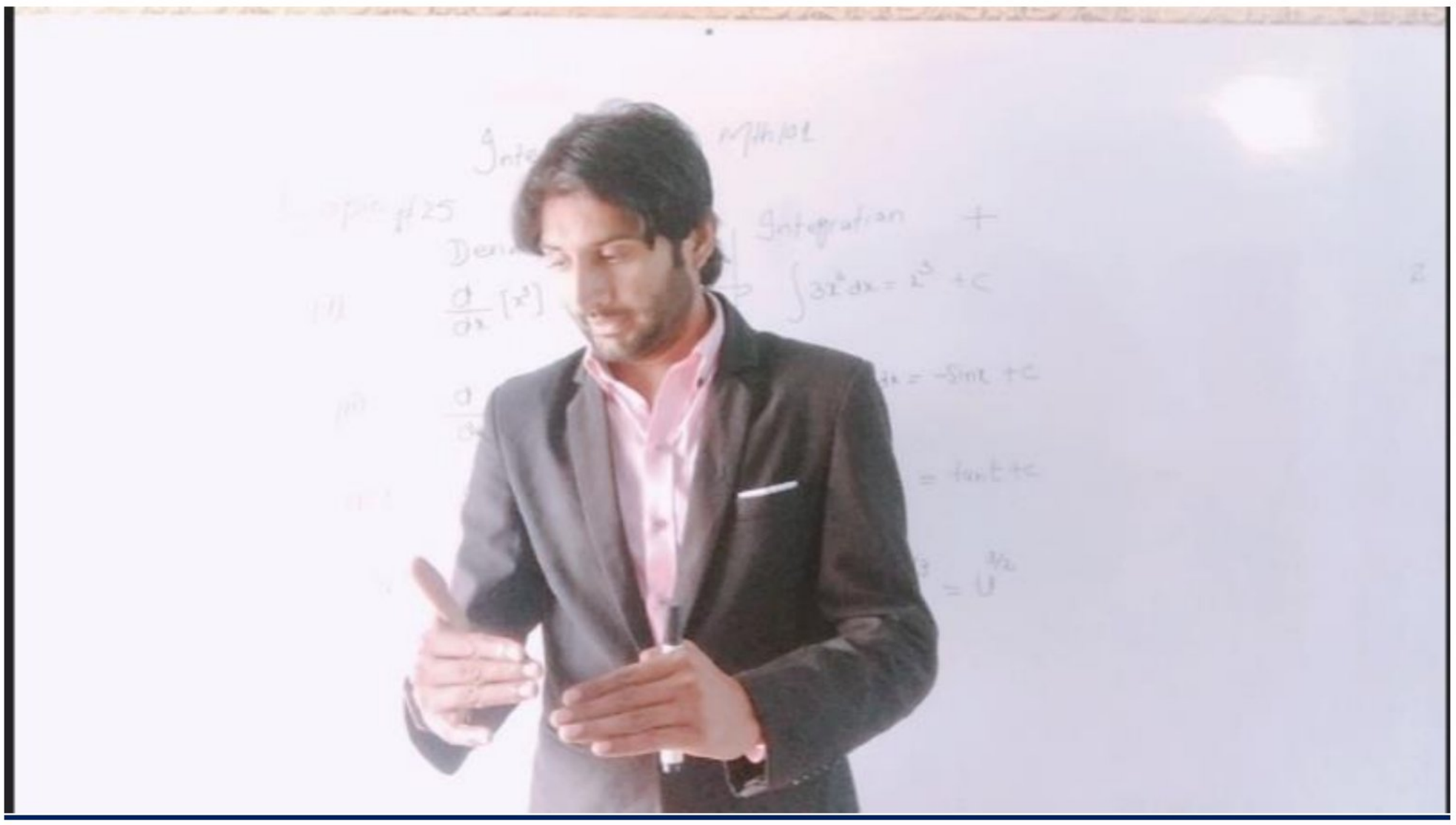
An improper integral is the limit of a definite integral as an endpoint of the interval of integration approaches either a specified real number or ∞ or -∞ or, in some cases, as both endpoints approach limits.

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TRUE Click here for detail

FALS

Take Care Good Bye



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